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Dynamic self-assembly of small molecules enables the spontaneous fabrication of hole conductors at perovskite/electrode interfaces for over 22% stable inverted perovskite solar cells†

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The bottom hole transport layers (HTLs) are of paramount importance in determining both the efficiency and stability of inverted perovskite solar cells (PSCs), however, their surface nature and properties strongly interfere with the upper perovskite crystallization kinetics and also influence interfacial carrier dynamics. In this work, we strategically develop a simple, facile and spontaneous fabrication method of the HTL at the perovskite/electrode interface by dynamic self-assembly (DSA) of small molecules during perovskite crystallization. Different from the traditional layer-by-layer approach, this DSA strategy involves a bilateral movement of self-assembled molecules (SAMs) from perovskite solution, realizing simultaneous fabrication of the HTL and perovskite surface passivation. We design a multifunctional molecule, (4-(7*H*-benzo[c]carbazol-7-yl)butyl)phosphonic acid (BCB-C4PA), for the DSA process, to optimize both self-assembly ability and interfacial energy alignment. Benefitting from this unconventional DSA approach and BCB-C4PA, a champion PCE of 22.2% is achieved along with remarkable long-term environmental stability for over 2750 h, which is among the highest reported efficiencies for SAM-based PSCs. This investigation provides a creative, unique and effective molecular approach for preparing reliable charge transport layers, opening up new avenues for the further development of efficient interfacial contacts for PSCs.

New concepts

In this work, we for the first time demonstrate a dynamic self-assembly (DSA) strategy for the simultaneous fabrication of a hole transport layer and modulation of perovskite crystallization in the solution processing of perovskite solar cells (PSCs). Such a one-stone-for-two-birds strategy can facilely construct a high-quality buried interface between the perovskite and the ITO electrode to reduce defect-associated recombination losses, which can hardly be achieved by the traditional stepwise spin-coating of the perovskite layer and hole transport layer. This approach highlights the development of suitable self-assembled molecules (SAMs) for bilateral movement to generate a nanometer-thick hole transport layer and promote perovskite surface passivation when processing in their mixture solution with the perovskite precursor. The SAM-based hole transport layer forms a pinhole-free and homogeneous interface to promote hole extraction from the perovskite. By adopting the DSA approach, perovskite solar cells can readily exhibit a significantly improved power conversion efficiency (>22.2%) along with exciting long-term environmental stability for over 2750 h. Importantly, our strategy can be a big advantage when up-scaling high-efficiency stable PSCs *via* slot-die coating or solution printing for large-area device applications, since it is almost impossible to achieve homogeneous and even nm-scale SAM interfacial layers through conventional step-by-step spin-coating.

Introduction

Photovoltaic (PV) technologies that convert solar energy directly into electricity have received worldwide attention and been entrusted with high expectations to substitute conventional non-renewable electricity production. Among new-generation PVs, emerging perovskite solar cells (PSCs) are recognized as one of the most promising frontrunners with the record power conversion efficiency (PCE) surpassing 25%, comparable to that of their commercial c-Si counterpart (>26%).[1] Moreover, the high capability of solution-processable fabrication adds additional benefits to this PSC technology for future commercialization.[2]

Inverted PSCs with a p–i–n configuration combine the advantages of easy fabrication, superior flexibility, negligible current–voltage hysteresis and high compatibility with tandem cells.[3] However, their photovoltaic performance lags behind the conventional n–i–p structured PSCs, which is mainly limited by the poor buried interfaces as well as severe defect-associated recombination losses. Therein, the hole transport layer (HTL)

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acting as the perovskite crystallisation substrate in inverted PSCs plays essential roles in determining the film quality and defect chemistry of perovskite thin films. Unfortunately, the state-of-the-art HTL poly[bis(4-phenyl) (2,4,6-trimethylphenyl) amine] (PTAA) exhibits poor surface hydrophobicity and wettability, significantly limiting the reproducibility of film preparation as well as the device performance. Although different strategies to circumvent these problems by finely tuning the molecular structure of HTLs or surface post-treatment have been reported, the involved material synthesis and/or sophisticated device fabrication increase the device complexity, which hinders large-scale commercial applications.[4–6] Therefore, searching for alternative HTLs with superior surface properties is of importance for optimizing the buried interfaces and enhancing the device performance.

Over the past few years, many types of HTLs targeting cost-competitive and high performance p–i–n PSCs have been developed for layer-by-layer spin-coating of devices.[7–10] Among them, self-assembled monolayers (SAMs) show great potential owing to their multiple advantages such as low material consumption, compatibility with flexible substrates, tunable bandgap, high transmittance and green-solvent processability.[11] Featuring anchoring groups (such as carboxylic acid or phosphonic acid),[12,13] they can spontaneously adsorb onto the surface of oxide substrates to form a densely packed and uniform monolayer coverage. Such a film with one to a-few-molecule thickness does change the work function (WF) of the contact interface, and facilitate the charge extraction from the perovskite layer if the above-deposited perovskites have suppressed interfacial defects and promoted crystallization dynamics.[14] These versatile merits endow SAM based PSCs with largely improved device efficiency and operational stability.[15–17] So far, various types of SAMs (*e.g.* phenothiazine, carbazole, triphenylamine derivatives, *etc.*) have been developed as HTLs for inverted PSCs, showing tunable energy levels and good recombination blocking effects, leading to high PCEs over 22% and promising stability.[11,12,16,18]

Unfortunately, such layer-by-layer processed SAMs are still not feasible for commercial mass production considering the following issues: (1) in principle, SAMs should be as thin as possible to minimize the resistive losses, whereas at the same time they should be dense and compact to block undesired charge recombination.[9] This demanding requirement is difficult to be met under the large area conditions and requires careful formulation of the solution; (2) the surface-wettability of the underlying SAMs can significantly affect the nucleation and growth of the upper perovskites, which makes it very challenging to fabricate highly crystalline and defect-free perovskite films on large-area substrates;[19] (3) currently, the fabrication of inverted PSCs highly relies on the nonscalable spin-coating process in the aspects of large material waste and low suitability for large film production.[20] Although all of these factors obviously increase the capital expenditure of manufacturing equipment and processing complexity, little attention has been paid to the development and optimization of thin-film fabrication processes.[21–23]

In this work, we propose a simple, facile and unique dynamic self-assembly (DSA) strategy for the spontaneous fabrication of HTLs by incorporating SAMs directly into perovskite precursor solutions. We discovered that bilateral movement of such small molecules from solution during perovskite crystallization could realize the simultaneous formation of HTLs and perovskite surface passivation. We strategically designed a multifunctional SAM, namely, 4-(7*H*-benzo[*c*]carbazol-7-yl)butyl)phosphonic acid (BCB-C4PA), which is grafted with a strong anchoring group and a large fused-ring aromatic moiety, in order to strengthen the adsorption ability and optimize the energy alignment. Structural analysis revealed that the HTL based on BCB-C4PA made by DSA substantially improves the microstructural uniformity of the buried interfaces and passivates the defects on the top surface, which favors enhancing the charge collection and suppressing non-radiative recombination losses. Consequently, the corresponding devices based on DSA and BCB-C4PA exhibited a remarkable PCE of 22.2%, which is much higher than those of the devices fabricated by traditional layer-by-layer spin coating (17.9%) and the previously reported [4-(9*H*-carbazol-9-yl)butyl]phosphonic acid (CZ-C4PA) based devices (18.2%) under the same conditions.[24] Moreover, benefitting from the improved perovskite quality and interfacial functionalization, the devices based on BCB-C4PA and DSA demonstrated outstanding long-term stability by retaining over 90% PCE even after 2750 h ambient storage. This work develops an effective, unique and generally applicable fabrication technique to prepare interfacial contact layers, and underscores the promising potential of DSA in preparing functional layers for various optoelectronic applications.

Results and discussion

Design and characterization of *in situ* self-assembled SAMs

The fabrication processes of HTLs by the DSA strategy and traditional layer-by-layer processing are schematically presented in Fig. 1a and b. During conventional spin coating, a SAM-based solution is firstly coated on conductive glass, followed by the deposition of a perovskite layer on top of the SAM layer. The interaction between the SAM and the perovskite precursors often induces the deformation of the SAM layer and the wetting issues of perovskite formation. In contrast, our developed DSA strategy only involves the one-step coating of the solution mixture of SAMs and perovskite precursors, which enables the simultaneous deposition of both the HTL and the perovskite film on the glass substrates.[25–27] To the best of our knowledge, the DSA method is the first demonstration of a one-pot deposition technique to realize the formation of several functional layers, whose working mechanism will be discussed later. This DSA technology not only simplifies the fabrication process, but also eliminates the substrate effects on perovskite crystallization.

In the design of *in situ* self-assembled SAMs, we focus on the following four key considerations, *i.e.*, (i) SAMs should have

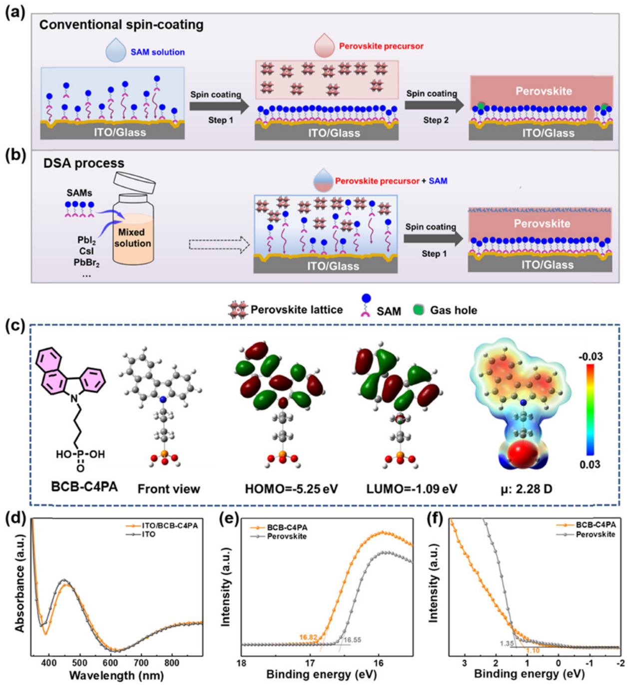

Fig. 1 Comparison of device fabrication processes using different strategies and the molecular properties of our designed SAM. Schematic illustration of the independent spin-coating method (a) and DSA strategy (b) for preparing the perovskite layer and a HTL with a SAM. The DSA strategy exhibits simplified preparation processes and an improved buried interface. (c) Chemical structure and molecular simulation (including molecular geometry, frontier molecular orbital energy levels, electrostatic potential surface and dipole moment) of the designed BCB-C4PA. (d) UV-vis absorbance spectra of ITO and ITO/BCB-C4PA films (0.5 mg mL^{-1} in ethanol). (e) and (f) Ultra-violet photoelectron spectra (UPS) of BCB-C4PA and perovskite films spin-coated on ITO substrates (excitation energy is 21.22 eV).

good solubility in perovskite precursor solvents (*e.g.* DMSO) in order to prepare homogeneous mixtures for solution processing; (ii) the anchoring groups in one end of SAM molecules secure strong covalent bonding with metal oxide substrates (*e.g.* indium tin oxide (ITO)) even at room temperature;[28] (iii) large fused-ring conjugated terminals in SAMs prevents them from fitting into the perovskite lattice and thus exerts a little negative effect on the formation of perovskite films;[17] and (iv) SAMs possess suitable energy-level alignment with both perovskite and metal oxide substrates for efficient interfacial charge transfer. Moreover, when such SAMs migrate down and adsorb onto the ITO substrate, they can also act as additives to promote the crystallization of perovskite and passivate the defects at the buried interfaces and grain boundaries.

Under these conditions, a small molecule BCB-C4PA was rationally designed by adopting an asymmetric conjugated moiety, *i.e.*, benzo[*c*]carbazole (BCB), to increase the solubility in common organic solvents (*ca.* ethanol, DMSO) and also the molecular dipole moment (μ) to facilitate electron delocalization.[29] A higher dipole moment can endow BCB-C4PA with superior ability to modify the WF of the ITO substrate, thereby realizing better energy alignment to the valence band maximum (VBM) of the perovskite.[30] In addition, the high electron density on the BCB may potentially reduce the generation of interfacial defects *via* enhancing BCB-C4PA's interaction with the perovskite.[31] On the other hand, the length of the alkyl chain is important for lateral interaction between molecules during the self-assembly process, which affects the final packing structure.[14] In this regard, a linkage of the C4 chain length is adopted instead of C2 to increase the molecular flexibility for optimum *in situ* self-assembly in SAM films.

To demonstrate the merits of the DSA strategy with BCB-C4PA for p–i–n PSCs, we have thus conducted a systematical study of its synthesis, structure characterization, molecular simulation and photovoltaic performance. The molecular structure of BCB-C4PA is presented in Fig. 1c, whose detailed synthesis, purification and structural characterization of the intermediates are given in the ESI.† The cheap raw materials and straightforward synthetic routes render a significant low-cost advantage for BCB-C4PA with laboratory synthetic costs of about 20 \$ g^{-1} (Table S1, ESI†), much lower than that of commercially available HTLs (*e.g.* PTAA, 1980 \$ g^{-1}).[32]

To gain insights into molecular geometries, frontier molecular orbital (FMO) energy levels, electrostatic potential surface (EPS) and dipole moment, molecular simulation was conducted using density functional theory (DFT) at the B3LYP level with 6-31G as the basis set (Fig. 1c).[33] BCB-C4PA has an estimated highest occupied molecular orbital (HOMO) energy level of −5.25 eV and a lowest unoccupied molecular orbital (LUMO) energy level of −1.09 eV. From EPS analysis, it presents electrostatic potential with a well-separated distribution along the head-to-tail molecular geometry. The BCB moiety with negative charge (red) delocalization may function as the Lewis base to interact with the inorganic $[PbI_6]^-$ framework and induce the formation of high-quality perovskite films.[31] Furthermore, the dipole moment of BCB-C4PA is calculated to be 2.28 D due to its high intrinsic polarity.

Subsequently, the optical properties of BCB-C4PA in both solution (Fig. S7a, ESI†) and film states (Fig. 1d) were characterized by UV-vis absorption spectroscopy. Its optical bandgap (E_g^{opt}) was estimated to be 3.26 eV while the film shows reduced absorption in the region of 390–460 nm compared to the bare ITO substrate (Fig. 1d). The ultraviolet photoelectron spectroscopy (UPS) measurements of the BCB-C4PA covered ITO substrate and the baseline perovskite $Cs_{0.07}FA_{0.9}MA_{0.03}Pb(I_{0.92}Br_{0.08})_3$ (hereafter labelled CsFAMA) were then performed to prove BCB-C4PA's suitable energetic properties as HTLs for PSCs. The WFs of BCB-C4PA covered ITO and CsFAMA were calculated to be −4.4 and −4.67 eV (Fig. 1e and f), respectively. With respect to the energy offsets between the perovskite VBM and the HOMO energy level of the studied BCB-C4PA, the value was calculated to be 0.5 eV. The lowered HOMO energy level of BCB-C4PA ensures sufficient driving force for hole extraction, and simultaneously its LUMO energy level is also shallow enough to block electron leakage. Since HTLs should possess good thermal stability in consideration of device fabrication and shelf-life stability of PSCs,[34] the thermal property of designed BCB-C4PA was measured *via* thermogravimetric analysis (TGA). As shown in Fig. S7b (ESI†), BCB-C4PA exhibited good thermal stability with a decomposition temperature (T_d, 5% weight loss temperature) approaching 235 °C.

DSA strategy with BCB-C4PA

BCB-C4PA associated with the DSA strategy to generate a high-quality interfacial hole conductor and perovskite buried interfaces is confirmed by X-ray photoelectron spectroscopy (XPS), time-of-flight secondary-ion mass spectroscopy (TOF-SIMS) and scanning electron microscopy (SEM) characterization. To probe the necessity of the phosphonic acid anchoring group as well as the effect of terminal units on self-assembly ability, the literature reported 9-(4-bromobutyl)-9*H*-carbazole (CZ-C4Br) and CZ-C4PA (their molecular structures are shown in Scheme S1, ESI†)) were also prepared for comparison.[24] The characteristics of CZ-C4PA including molecular simulation, energetic properties and optical absorption are summarized in Fig. S8 (ESI†).

The molecules with phosphonic acid anchoring groups can form robust covalent bonds on ITO substrates by reacting with the surface hydroxyl groups.[11,35] This may be beneficial for SAM spontaneous migration towards ITO substrates to form a specific monolayer. To this end, XPS was first performed to analyze the atomic bonds of studied HTLs on ITO substrates. As there is no perovskite component, they are dissolved in isopropanol solvent and spin-coated directly on ITO substrates. Fig. S9 (ESI†) shows the XPS spectra of the In 3d binding energy region for samples. Apparently, remarkable shifts of ~2 eV to higher binding energies were observed from pure ITO to ITO/BCB-C4PA and ITO/CZ-C4PA films, while for the ITO/CZ-C4Br sample a negligible shift was observed. These confirm the existence of strong interactions between BCB-C4PA (CZ-C4PA) and the ITO substrates, in line with the previous results.

Afterward, we added these SAMs into perovskite precursor solution and spin-coated the mixtures on ITO substrates. The resulting perovskite films were detected by TOF-SIMS to explore the spatial distribution of incorporated molecules.[36] The signal of PO_3^- originates from BCB-C4PA and CZ-C4PA, while the I_2^- and $In_2O_2^-$ ions are assigned to the perovskite layer and ITO substrate, respectively.[37] The depth profiles with the corresponding 3D distribution pictures are presented in Fig. 2a–d. As illustrated, PO_3^- is only distributed on the top and bottom surfaces of the perovskite film in both BCB-C4PA and CZ-C4PA incorporated samples, which means they are separated from the perovskite domain during the perovskite crystallization through the upward and downward migration, as illustrated in Fig. 1b. A part of them remained on the top surface of the perovskite film could function as the passivation layer, while the other sinking down towards ITO substrates could spontaneously form a compact HTL. Notably, the contents of BCB-C4PA and CZ-C4PA existing on the top surface is very low as the high counts of PO_3^- only lasted for the first few seconds (at a sputtering rate of 10 nm min^{-1}). As for the CZ-C4Br sample, the molecular weight signal (302 g mol^{-1}) indicating its distribution was not detected and no intermediate layer existed between the perovskite layer and the ITO substrate (Fig. S10, ESI†). We speculated that it was due to its low concentration after dispersing in the perovskite film. This highlights the important role of phosphonic acid as the anchoring group for the DSA strategy.

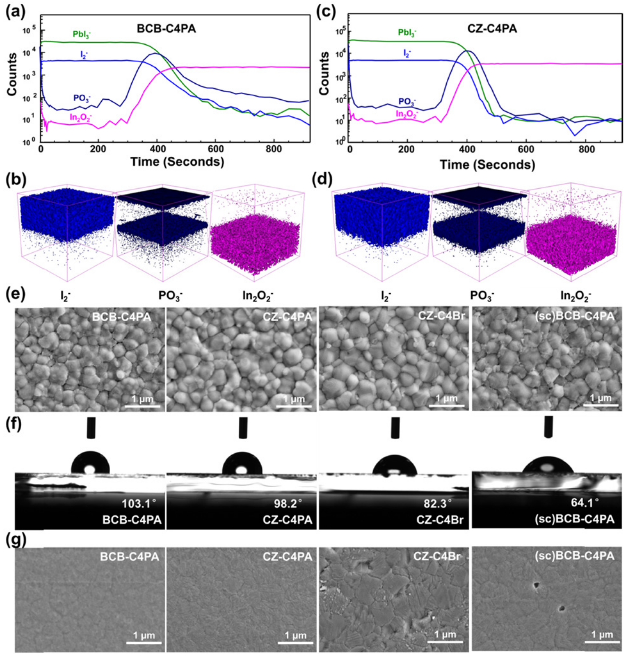

Fig. 2 The space distribution of incorporated SAMs using the DSA strategy and their effects on perovskite top/buried interfaces. TOF-SIMS profiles and the corresponding 3D distribution pictures of perovskite films prepared by the DSA strategy using BCB-C4PA (a and b) and CZ-C4PA (c and d). Top-view SEM images (e), contact angle measurements (f) and buried interface SEM images (g) of perovskite films prepared by the DSA strategy using BCB-C4PA, CZ-C4PA and CZ-C4Br. For comparison, the perovskite film deposited on independently spin-coated BCB-C4PA, coded as (sc)BCB-C4PA, was also investigated.

We then performed X-ray diffraction (XRD) and SEM analyses to reveal the effects of the *in situ* formed HTLs on the properties of the bulk and the top surface of the perovskite film. For comparison, the HTL made by conventional spin-coating of BCB-C4PA, denoted as (sc)BCB-C4PA, was also investigated for reference. The results showed negligible differences in the X-ray diffractograms between BCB-C4PA (DSA) and (sc)BCB-C4PA samples, suggesting that *in situ* prepared BCB-C4PA does not significantly change the crystallization of CsMAFA (Fig. S11, ESI†). Fig. 2e displays the top-view SEM images of different films, and interestingly some organic thin layers can be seen on the surface and grain boundaries of perovskite films based on BCB-C4PA and CZ-C4PA, which are consistent with the TOF-SIMS results. We further conducted contact angle measurements to investigate the surface hydrophobicity of different perovskite films. The water droplet contact angle on the (sc)BCB-C4PA based perovskite film (64.1°) was close to the value of pristine perovskite (63.2°), whereas it remarkably increased to 98.2° and 103.1° on the perovskite surfaces with CZ-C4PA and BCB-C4PA, respectively (Fig. 2f). This implies that the presence of SAMs improves the hydrophobicity of perovskite films, which is beneficial for preventing external moisture intrusion to enhance device stability. The UV-vis absorption measurements of perovskite films with or without SAM-based HTLs were almost the same (Fig. S12, ESI†), suggesting little effect on the optical properties of perovskite films.

Aside from the top surface investigation, we further implemented a non-destructive technique to integrally exfoliate the SAM-containing perovskite films from ITO substrates,[38] thereby directly visualizing their buried interfaces. The specific experimental details are provided in the ESI.† As displayed in Fig. 2g, the grain boundaries on the bottom surface of the CZ-C4PA sample becomes more blurred relative to the pure perovskite film (Fig. S13b, ESI†), which may suggest the formation of a CZ-C4PA organic layer beneath the perovskite crystals. In the case of BCB-C4PA, this phenomenon is even more pronounced, demonstrating that the self-assembled BCB-C4PA layer could be denser than the CZ-C4PA one. We speculate that this may be due to the larger molecular size (weight), and/or the larger dipole moment of BCB-C4PA. However, in the case of CZ-C4Br, the grain boundaries are still very clear. These results confirm again that SAMs form a uniform network adherent to the interface between ITO and the perovskite layer. Notably, the bottom surface morphologies of BCB-C4PA and CZ-C4PA based perovskite films are uniform and void-free, much better than that of the (sc)BCB-C4PA counterpart. This is beneficial for reducing the defect concentration at the bottom interface and preventing direct contact between the perovskite and the electrode, thus improving the device efficiency and stability.[39] Fig. S13c and d (ESI†) show the pictures of perovskite films fabricated by BCB-C4PA (DSA) and (sc)BCB-C4PA, in which the former is much more uniform, showing the advantage of the DSA strategy to eliminate the negative effect of HTL's surface wettability.

Device performance and mechanism analysis

According to the above characterization results, we fabricated inverted PSCs with a configuration of ITO/SAMs + CsMAFA (E_g = 1.59 eV)/C60/BCP/Ag (Scheme S2, ESI†), in which SAMs were incorporated in the perovskite precursors and then directly spin-coated on the ITO substrates. (sc)BCB-C4PA and CZ-C4Br based devices were also constructed as reference samples. More fabrication details are provided in the ESI.†Fig. 3a illustrates the current density–voltage (J–V) curves of the best-performing PSCs with BCB-C4PA, CZ-C4PA, CZ-C4Br and (sc)BCB-C4PA as HTLs, and the corresponding photovoltaic parameters are summarized in Table 1. The performance parameters of optimization processes for different devices are summarized in Fig. S15–S17 (ESI†). As demonstrated, the device employing CZ-C4PA and (sc)BCB-C4PA delivered moderate PCEs of 18.2% and 17.9%, respectively. In contrast, the CZ-C4Br samples yielded a very low PCE (6.6%). Encouragingly, the devices with *in situ* incorporated BCB-C4PA exhibited a champion PCE of 22.2% with a J_{SC} of 24.4 mA cm^{-2}, a V_{OC} of 1.13 V, and a FF of 80%, outperforming most of the SAM based PSCs reported so far (Fig. 3b). Fig. 3c displays the corresponding external quantum efficiency (EQE) spectra for the devices based on different HTLs. The integrated J_{SC} values were calculated to be 23.9, 21.9, 14.1 and 22.1 mA cm^{-2} for BCB-C4PA, CZ-C4PA, CZ-C4Br and (sc)BCB-C4PA, respectively,

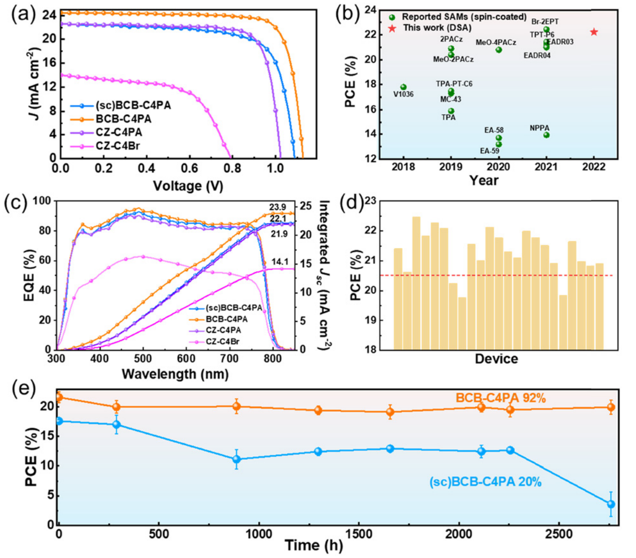

Fig. 3 Photovoltaic performance and long-term stability of DSA-based solar cells with a triple cation perovskite absorber. (a) J–V curves of the best cells using *in situ* incorporated BCB-C4PA, CZ-C4PA, and CZ-C4Br. To demonstrate the superiority of the DSA strategy, the complete device with independently spin-coated BCB-C4PA was also prepared. (b) PCE value achieved by the DSA strategy using BCB-C4PA in comparison to that of the reported independently spin-coated SAM.[40] (c) The corresponding EQE spectra of devices with *in situ* incorporated BCB-C4PA, CZ-C4PA, CZ-C4Br and independently spin-coated BCB-C4PA. (d) PCE histograms of 23 devices prepared by the DSA strategy using BCB-C4PA. (e) Stability assessments of PSCs prepared by the DSA strategy using BCB-C4PA under a N_2 atmosphere at room temperature.

Table 1 Photovoltaic parameters of PSCs with *in situ* incorporated BCB-C4PA, CZ-C4PA, CZ-C4Br and independently spin-coated BCB-C4PA under AM 1.5 G illumination at 100 mW cm^{-2}

HTL	Devices	J_{sc} [mA cm^{-2}]	V_{oc} [V]	FF [%]	PCE [%]
BCB-C4PA	Forward scan	24.4	1.13	80	22.2
	Reverse scan	24.5	1.12	79	21.8
CZ-C4PA	Forward scan	22.6	1.02	79	18.2
	Reverse scan	22.6	0.97	75	16.4
CZ-C4Br	Forward scan	14.0	0.79	59	6.6
	Reverse scan	14.2	0.67	42	4.0
(sc)BCB-C4PA	Forward scan	22.6	1.09	73	17.9
	Reverse scan	22.8	1.08	72	17.7

which matched well with the J–V measurement results. Furthermore, devices using *in situ* incorporated BCB-C4PA exhibited good reproducibility with more than 85% of the devices (20 devices) having PCEs higher than 20.5% (Fig. 3d). To verify the generality of the DSA strategy, we also fabricated another perovskite system (E_g = 1.55 eV) based on PSCs using the DSA method and a high PCE of 22.49% is achieved as shown in Fig. S18 (ESI†), indicating that DSA is a generally applicable fabrication technique. Device stability is another important factor influencing the reliability of such SAM materials. Therefore, the long-term stability of these devices was examined by storing the devices under N_2 at room temperature, as shown in Fig. 3e. According to the time evolution of PCEs for PSCs, the unencapsulated device based on BCB-C4PA (DSA) maintained >90% of the initial PCE after storage for over 2750 h, whereas the PCE of CZ-C4PA and (sc)BCB-C4PA based PSCs declined to 75% and 20% over the same period (Fig. S19, ESI†). These results further proved that BCB-C4PA with the DSA strategy favours the enhancement of device stability.

To explore the origins of the device performance variation, we first performed the Mott–Schottky (MS) analysis on devices to evaluate the built-in potential (V_{bi}) and charge distribution at the HTL/perovskite interface (Fig. 4a).[41] The V_{bi} of the devices was obtained from the interceptions of the linear region with the x-axis. The V_{bi} values were estimated to be 1.10 V, 0.99 V, 0.87 V and 1.10 V for BCB-C4PA, CZ-C4PA CZ-C4Br and (sc) BCB-C4PA, respectively. The larger V_{bi} value of the BCB-C4PA device signifies a stronger driving force for photogenerated charge carrier separation, transport, and collection.[42] In addition, the carrier density at the HTL/perovskite interface can be estimated by analyzing the slope of the Mott–Schottky plots. The slope of PSCs with BCB-C4PA is much higher than those with CZ-C4PA, CZ-C4Br and (sc)BCB-C4PA, demonstrating faster charge transfer at the HTL/perovskite interface with reduced carrier accumulation in the BCB-C4PA based sample.

We then performed electrochemical impedance spectroscopy (EIS) to articulate the carrier transport and recombination resistance in PSCs. The corresponding values are summarized in Table S2 (ESI†). Generally, the semicircle in the low-frequency region represents the recombination resistance (R_{rec}) while the high-frequency region is related to transport resistance (R_{ct}).[43] As shown in Fig. 4b, the BCB-C4PA based device exhibited a lower R_{ct} (26.7 Ω) and a larger value of R_{rec} (1.59 kΩ) than other samples. This indicates that *in situ* incorporated BCB-C4PA can effectively suppress charge-carrier recombination and improve charge transfer, which is beneficial for enhancing V_{OC} and the FF. Dark current analysis was conducted to further elucidate the leakage current formed by carrier recombination in our PSCs. As shown in Fig. 4c, the BCB-C4PA based device exhibited a much lower leakage current in contrast to the other counterparts. This implies again that the charge-carrier recombination process is reduced and more photocurrents would flow through the device instead of direct shunting.[44]

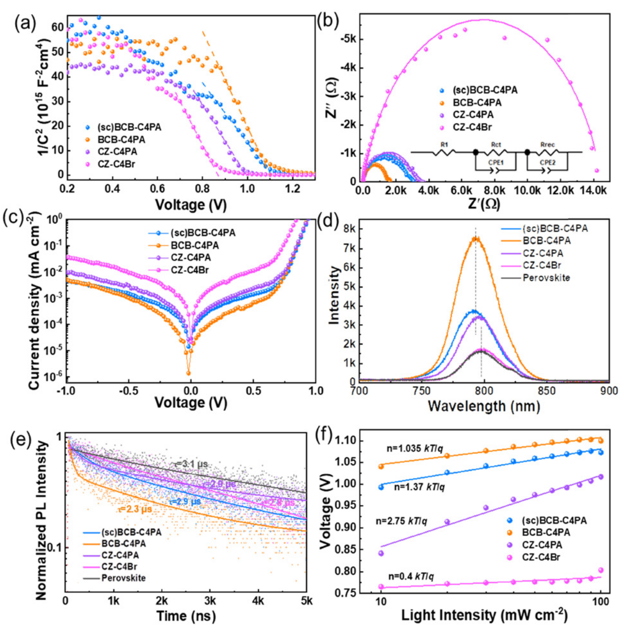

Fig. 4 Interfacial charge transport and recombination in PSCs prepared by the DSA strategy and independent spin-coating method. (a) Mott–Schottky plots and (b) EIS for PSCs with the respective HTLs. (c) J–V curves of PSCs measured under dark conditions of the respective HTLs. (d) Steady-state PL and (e) TRPL transient spectra of CsMAFA films with or without respective HTLs. The TRPL curves were fitted using a biexponential decay function to obtain the decay time values. (f) Intensity-dependent V_{OC} with linear fits (solid line) of PSCs with the respective HTLs.

Steady-state photoluminescence (PL) and time-resolved photoluminescence (TRPL) studies were further performed to scrutinize the charge transfer kinetics at the HTL/perovskite interface. As indicated in Fig. 4d, the PL intensity increased sharply for BCB-C4PA incorporated perovskites compared with CZ-C4PA and (sc)BCB-C4PA ones, indicating significantly suppressed non-radiative recombination.[45] Besides, the PL peak of the film blue-shifted from 790 to 780 nm after incorporating BCB-C4PA, which may be attributed to reduced defects at grain boundaries and/or interfaces (top surface and the buried interface of perovskites).[42] Fig. 4e illustrates the TRPL decay curves of different perovskite films on inert glass substrates with the corresponding data of carrier lifetimes presented in Table S3 (ESI†). The TRPL decay curves were fitted using a biexponential decay function:

$$\tau = \frac{A_1\tau_1{}^2 + A_2\tau_2{}^2}{A_1\tau_1 + A_2\tau_2}$$

where τ_1 and τ_2 are the short and long lifetime constants, respectively, while A_1 and A_2 are their corresponding decay amplitudes.[46] As elaborated in the literature, τ_1 is related to the charge transfer to the HTL, whereas τ_2 arises from the interface non-radiative recombination losses.[47] The results showed that BCB-C4PA incorporated perovskite exhibits a sharply shortened τ_1 of 0.09 μs, an increased portion of A_1, and ultimately a lower average lifetime (τ_{avg}) of 2.3 μs in relative to the CZ-C4PA counterpart (τ_{avg} = 2.8 μs). This means that the BCB-C4PA incorporated device has superior charge transfer ability to mitigate the negative effect of charge recombination which finally contributes to the high FF value. These results are consistent with the PL quenching studies, implying more efficient charge transport and suppressed non-radiative recombination losses occurring in the BCB-C4PA device.[23,48]

To gain a deeper understanding of the charge recombination kinetics in different PSCs, the V_{OC} dependence on the illumination intensity (P) was examined by varying the incident light intensity from 10 to 100 mW cm^{-2}. The V_{OC}–P curves are plotted in Fig. 4f using the following equation $V_{OC} \propto nkT/q \ln P$, in which n, q, k and T are the ideality factor, elementary charge, Boltzmann's constant and temperature, respectively.[49] Generally, the slope of the V_{OC}–P curves close to $2kT/q$ implies a greater probability of trap-assisted recombination.[50] As presented, the slope was reduced from 2.75 kT/q for the CZ-C4PA based sample to 1.035 kT/q for the BCB-C4PA based device, which suggests that trap-assisted recombination was successfully reduced.

Ultimately, we employed proton nuclear magnetic resonance (^{1}H NMR) to gain insight into the interfacial interaction between BCB-C4PA and the perovskite, which is highly related to defect passivation.[5] The uncoordinated Pb^{2+} defects at grain boundaries and surfaces are reported as the main defect sources in perovskites, and thus PbI_2 was adopted to mimic undercoordinated Pb^{2+} ions in the perovskite films and bent with BCB-C4PA solution.[51] As shown in Fig. S20 (ESI†), evident down-field shifts of proton peaks as well as broadened signals (at 8.0 and 7.5 ppm) were observed in BCB-C4PA samples after blending with PbI_2, suggesting the redistribution of the electron density along the molecular backbone caused by coordinating interactions between BCB-C4PA and the Pb^{2+} ions.[8] Based on these results, we can draw a conclusion that BCB-C4PA located on the top and bottom surfaces of perovskite can manipulate the defects so as to minimize interfacial non-radiative recombination.[5]

Conclusions

In summary, we have demonstrated a simple, facile and unique dynamic self-assembly strategy (DSA) technique to simultaneously deposit the perovskite layer and HTL in an effective one-step process. The results indicate that the SAMs added to the perovskite precursor solution can spontaneously migrate toward the ITO surface to form a distinct HTL during spin coating; meanwhile a small amount of them diffuses to the top surface of the perovskite film to passivate the defects at the surface and grain boundary. Consequently, the DSA strategy does not only improve the morphological uniformity of the perovskite layer, but also enhance charge transport and suppress non-radiative recombination. The champion device based on *in situ* incorporated BCB-C4PA yielded a champion PCE of 22.2%, which is among the highest PCE values of SAM based PSCs. In contrast, much lower PCEs of 17.9% and 18.2% have been obtained for (sc)BCB-C4PA and the previously reported CZ-C4PA devices, suggesting the positive and important roles of DSA in improving the device performance. More encouragingly, BCB-C4PA based PSCs without encapsulation also exhibited superior environmental stability by maintaining over 90% of their initial PCE even after 2750 h of storage. This work provides alternative ways to fabricate efficient HTLs for inverted solar cells, and the DSA strategy is generally applicable in the scalable deposition of transport layers in optoelectronic devices.

Experimental section

Synthesis of 7-(4-bromobutyl)-7*H*-benzo[*c*]carbazole (BCB-C4Br)

A mixture of BCB (2.00 g, 9.21 mmol), tetrabutylammonium bromide (TBAB) (0.59 g, 1.84 mmol), 1,4-dibromobutane (22.0 mL, 184.10 mmol) and 50% KOH aqueous solution (5.1 mL, 46.03 mmol) was stirred at 60 °C overnight. After completion of the reaction, the resulting solution was poured into water and extracted with dichloromethane. The combined extracts were dried over anhydrous magnesium sulphate, and the solvent was removed in a vacuum to afford the crude product. The crude product was then purified by column chromatography (silica gel, petroleum ether/dichloromethane, v/v, 4/1) to afford BCB-C4Br as a colourless oil (2.50 g, 77%). ^{1}H NMR (500 MHz, $CDCl_3$): 8.80 (d, J = 8.3 Hz, 1H), 8.61 (d, J = 7.9 Hz, 1H), 8.02 (d, J = 8.1 Hz, 1H), 7.92 (d, J = 8.8 Hz, 1H), 7.72 (t, J = 7.6 Hz, 1H), 7.66 (d, J = 8.8 Hz, 1H), 7.56 (d, J = 8.1 Hz, 1H), 7.49 (m, J = 14.6, 7.4 Hz, 2H), 7.40 (t, J = 7.5 Hz, 1H), 4.51 (t, J = 7.0 Hz, 2H), 3.38 (t, J = 6.5 Hz, 2H), 2.12 (m, J = 7.2 Hz, 2H), 1.93 (m, J = 6.7 Hz, 2H). ^{13}C NMR (125 MHz, $CDCl_3$): 139.36, 138.02, 130.19, 129.40, 129.10, 127.51, 127.19, 124.35, 123.76, 123.35, 123.09, 122.41, 120.06, 115.18, 110.76, 109.38, 42.40, 33.27, 30.35, 28.29.

Synthesis of diethyl (4-(7*H*-benzo[*c*]carbazol-7-yl)butyl) phosphonate (BCB-C4P)

7-Octyl-7*H*-dibenzo[*c*,*g*]carbazole (4 g, 10.54 mmol) was dissolved in a mixture of chloroform and acetic acid (1 : 1, 20 mL). After cooling in an ice-water bath, the reaction mixture was added with NBS (1.8 g, 31.62 mmol) dropwise in darkness. After the reaction overnight, the mixture was diluted with water and extracted with petroleum ether (100 mL). The organic phase was washed with water (2 × 50 mL) and brine (2 × 50 mL) before drying over anhydrous Na_2SO_4. The solvents were removed with a reduced vacuum. The yellow residue was purified by column chromatography over silica gel (petroleum ether as the eluent) to give the title compound **2** as a white powder (5.27 g, 93%). ^{1}H NMR (500 MHz, $CDCl_3$, δ): 9.12 (d, 2H), 8.47 (d, 2H), 8.04 (s, 2H), 7.71 (t, 2H), 7.65–7.57 (m, 2H), 4.47 (t, 2H), 1.94 (m, 2H), 1.44–1.17 (m, 10H), 0.86 (t, 3H). ^{13}C NMR (125 MHz, $CDCl_3$, δ): 137.00, 129.84, 128.58, 127.79, 126.46, 125.30, 124.80, 121.02, 117.14, 115.33, 43.67, 31.99, 30.09, 29.49, 29.36, 27.32, 22.82, 14.27.

Synthesis of (4-(7*H*-benzo[*c*]carbazol-7-yl)butyl)phosphonic acid (BCB-C4PA)

To a solution of BCB-4P (2.00 g, 4.88 mmol) in anhydrous 1, 4-dioxane (15 mL), bromotrimethylsilane (6.45 mL, 48.84 mmol) was added dropwise under a nitrogen atmosphere. After stirring for 24 h at room temperature, methanol (3 mL) was added and stirred for another 3 h. Finally, distilled water was added dropwise (25 mL), until the solution became opaque. The product was filtered off and washed with water to afford BCB-C4PA as a white solid (1.13 g, 65%). ^{1}H NMR (500 MHz, DMSO-d_6): 8.79 (d, J = 8.3 Hz, 1H), 8.62 (d, J = 7.9 Hz, 1H), 8.08 (d, J = 8.0 Hz, 1H), 8.02–7.94 (m, 2H), 7.79 (d, J = 8.2 Hz, 1H), 7.71 (t, J = 7.5 Hz, 1H), 7.48 (m, J = 6.8 Hz, 2H), 7.35 (t, J = 7.5 Hz, 1H), 4.57 (t, J = 7.1 Hz, 2H), 1.90 (m, J = 7.1 Hz, 2H), 1.61–1.49 (m, 4H). ^{13}C NMR (125 MHz, DMSO-d_6): 138.91, 137.74, 129.31, 129.20, 128.49, 127.14, 127.10, 124.19, 122.87, 122.82, 122.59, 121.79, 119.79, 113.75, 111.72, 110.11, 42.17, 30.18, 30.06, 27.85, 26.76, 20.45, 20.42. MALDI-TOF MS (m/z): $[M + H]^+$ calcd for $C_{20}H_{20}NO_3P$, 353.1181; found, 353.1171.

Device fabrication

ITO-coated glass substrates were sequentially sonicated in deionized water, glass cleaning solution, acetone and isopropanol, and then dried with dry compressed air. The cleaned ITO glass was treated with UV-ozone for 15 min to enhance wettability. For the preparation of the $Cs_{0.07}FA_{0.9}MA_{0.03}Pb(I_{0.92}Br_{0.08})_3$ precursor solution, 191.4 mg of FAI, 13.5 mg of FABr, 24.6 mg of CsI, 31.9 mg of MACl, 39.6 mg of $PbBr_2$, 0.64 mg of MAI and 591.2 mg of PbI_2 were added to 1 mL of DMF/DMSO (v/v,1 : 4) solution. And then, the SAM derivatives (BCB-C4PA, CZ-C4PA, CZ-C4Br) were mixed with perovskite precursor solutions at different molar concentrations of 1.8, 2.8, 3.7, 2.0, and 4.6 M. Moreover, the BCB-C4PA containing ethanol solution was spin-coated from the perovskite precursor solutions of different concentrations on the ITO substrate: 0.1, 0.3, 0.5, and 0.7 mg mL^{-1}. The perovskite solutions with and without SAM derivatives were filtered through a 0.22 μm polytetrafluoroethylene (PTFE) filter. 40 μL of BCB-C4PA containing perovskite precursor was directly spin coated on the ITO substrate through two step processes, first at 1000 rpm for 10 s and then at 4000 rpm for 30 s, and then 260 μL of EA was dropped on the spinning substrate during the second spin-coating step at 10 s before the end of the procedure. The substrate was then immediately transferred to a hot plate and was heated at 100 °C for 40 min under ambient air conditions with around 30% relative humidity. The BCB-C4PA-free perovskite solution was spin-coated on the BCB-C4PA-coated ITO glass in the same way as above. In the end, 30 nm thick C60 (Nano-C), 6 nm BCP and 80 nm Ag electrodes were thermally evaporated on the prepared perovskite film in a sequential order (device architectures are shown in Scheme S2, ESI†).

Conflicts of interest

There are no conflicts to declare.

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